

IN3140 — Obligatorisk oppgave 1

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Task 1

1a)

Finding the transformation matrix expressing the position and orientation of T with respect to B (T_T^B):

$$T_T^B = \begin{bmatrix} R_T^B & d_t^B \\ 0 & 1 \end{bmatrix}$$

Finding the rotation matrix R_T^B :

$$R_T^B = \begin{bmatrix} c_{\frac{\pi}{2}} & -s_{\frac{\pi}{2}} & 0 \\ s_{\frac{\pi}{2}} & c_{\frac{\pi}{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Finding the translation matrix d_T^B :

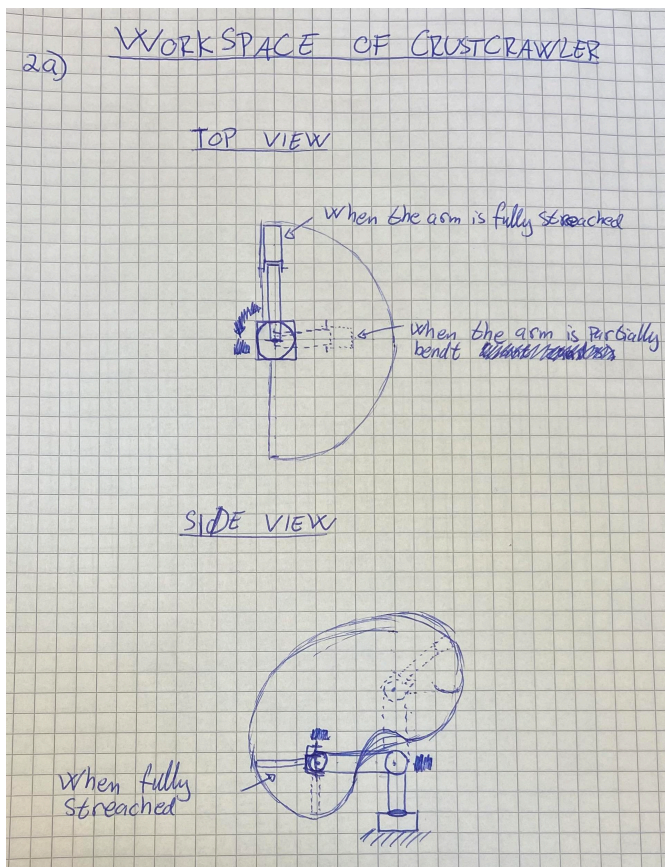
- Translation in x_B -direction is defined as $-y_w$ -direction (since they are parallel and pointed in opposite direction). So the translation from O_B to O_T in x-direction relative to coordinate system B is based in the y-coordinates in Figure 2. And therefore the translation from B to T in y-direction is: $-(400 - 650) = \underline{250}$
- In the same way we can see that translation in y_B direction correspond to $-x_w$. So translation from O_B to O_T in y-direction is: $-(1000 - 250) = \underline{-750}$
- The translation in z_B -direction is correspond to $-z_w$ -direction. So the translation from O_B to O_T in z-direction is: $-(900 - 1000) = \underline{100}$

$$d_T^B = \begin{bmatrix} 250 \\ -750 \\ 100 \end{bmatrix}$$

The total homogeneous transformation:

$$T_T^B = \begin{bmatrix} 0 & -1 & 0 & 250 \\ 1 & 0 & 0 & -750 \\ 0 & 0 & 1 & 100 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

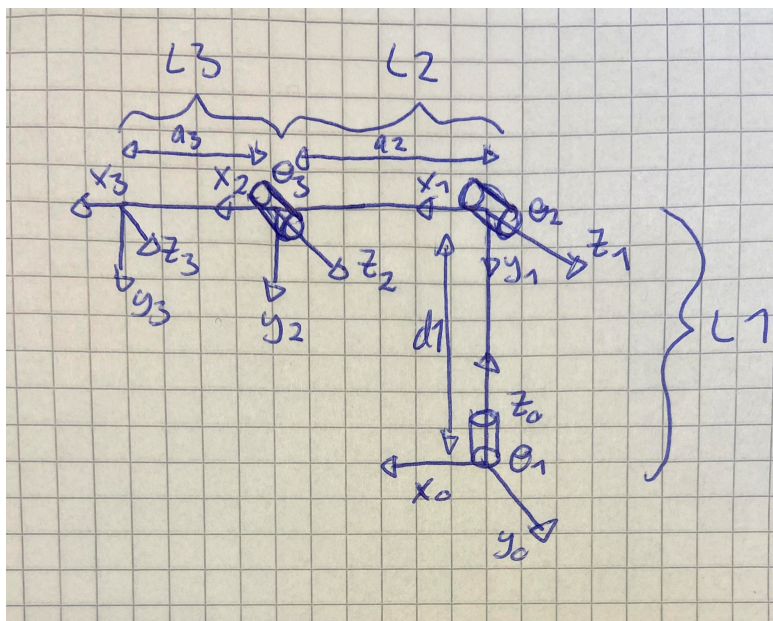
Task 2



2a)

I limited the joints to make clear how it affected the workspace.

- θ_1 is limited to approximately 180°
- θ_2 is limited to approximately 90°
- θ_3 is limited to approximately 350°



2b)

DH parameters:

	a_i	α_i	d_i	θ_i
L1	0	-90°	d_1	θ_1^*
L2	a_2	0	0	θ_2^*
L3	a_3	0	0	θ_3^*

* means variable

Chose to place the origin at the base of the first joint, at $0_0x_0y_0z_0$. Then placed z_0 to z_3 at the rotational axis of the three revolute joints. Then chose the x_0 arbitrarily, and placed y_0 according to the right hand rule. Chose x_1 to point in the direction of link 2 and parallel to x_0 (and perpendicular to both z_0 and z_1 to follow the convention). Lastly i chose the $0_2x_2y_2z_2$ - and $0_3x_3y_3z_3$ -coordinate system to match the previous so the numbers in the DH-table would be as easy as possible (no rotations). Then i filled in the missing y-axes.

2c)

From here c_i signify $\cos(\theta_i)$, c_2 signify $\cos(\theta_2)$ and c_3 signify $\cos(\theta_3)$. And the same applies to s_i , where $1 \leq i \leq 3$

Calculated matrix:

$$\begin{aligned}
 T_1 T_2 T_3 &= \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_1 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_3 & -s_3 & 0 & a_3 c_3 \\ s_3 & c_3 & 0 & a_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= (T_1 T_2) T_3 = \begin{bmatrix} c_1 c_2 & -s_2 c_1 & -s_1 & a_2 c_1 c_2 \\ s_1 c_2 & -s_1 s_2 & c_1 & a_2 s_2 + d_1 \\ -s_2 & -c_2 & 0 & -a_2 s_2 + d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_3 & -s_3 & 0 & a_3 c_3 \\ s_3 & c_3 & 0 & a_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= T_t^B = \begin{bmatrix} c_1 c_2 c_3 - c_1 s_2 s_3 & -c_1 c_2 s_3 - c_1 s_2 c_3 & -s_1 & a_3 c_1 c_2 c_3 - a_3 c_1 s_2 s_3 + a_2 c_1 c_2 \\ s_1 c_2 c_3 - s_1 s_2 s_3 & -s_1 c_2 s_3 - s_1 s_2 c_3 & c_1 & a_3 s_1 c_2 c_3 - a_3 s_1 s_2 s_3 + a_2 s_1 c_2 \\ -s_2 c_3 - c_2 s_3 & s_2 s_3 - c_2 c_3 & 0 & -a_3 s_2 c_3 - a_3 c_2 s_3 - a_2 s_2 + d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

Task 3

Filling inn the angles:

- $\phi_1 = 270^\circ$ gives $\theta_1 = 270^\circ$ since the coordinatesystem is the same
- $\phi_2 = 60^\circ$ gives $\theta_2 = -90^\circ + 60^\circ = -30^\circ$ because my coordinatesystem is angled 90° more in a counter clockwise direction than the configuration in the assingment (which is straight upwards)
- $\phi_3 = 45^\circ$ gives $\theta_3 = 45^\circ$ since when θ_2 is straightened upwards, link 2 and 3 will already be angled straight upwards (in z_0 -direction), wich makes it equal to the configuration in the assingment before the rotations

Filling inn the translations:

- L1 gives $d_1 = 100.9\text{mm}$
- L2 gives $a_2 = 222.1\text{mm}$
- L3 gives $a_3 = 136.2\text{mm}$

Coordinates of point p with respect to coordinate frame B:

$$p^B = T_t^B * p^t = \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & -227.6 \\ r_{31} & r_{11} & r_{33} & 343.5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -323.9 \\ 176.7 \\ 1 \end{bmatrix}$$

Coordinates of point p with respect to coordinate frame T:

$$p^T = T_B^T * p^B = (T_T^B)^{-1} p^B = \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix} p^B = \begin{bmatrix} 0 & 1 & 0 & 750 \\ -1 & 0 & 0 & 250 \\ 0 & 0 & 1 & -100 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -323.9 \\ 176.7 \\ 1 \end{bmatrix} = \begin{bmatrix} 426.1 \\ 250 \\ 76.7 \\ 1 \end{bmatrix}$$

So the coordinates of point p in coordinatesystem T (Task) is:

$$p^T = \underline{(426.1, 250, 76.7)}$$

Task 4

4a)

- **Analytical approach:** You base the x_c -, y_c - and z_c -coordinates on the d-column in the homogeneous transformation matrix (the translation). And from there you find expressions for each of the variables.
- **Geometrical approach:** You view the robot in different planes (two dimensions at a time). And calculate the variables that vary in that plane.

I will be using the analytical approach in task 4b)

4b) Analytical

$$T_t^B = \begin{bmatrix} c_1 c_2 c_3 - c_1 s_2 s_3 & -c_1 c_2 s_3 - c_1 s_2 c_3 & -s_1 & a_3 c_1 c_2 c_3 - a_3 c_1 s_2 s_3 + a_2 c_1 c_2 \\ s_1 c_2 c_3 - s_1 s_2 s_3 & -s_1 c_2 s_3 - s_1 s_2 c_3 & c_1 & a_3 s_1 c_2 c_3 - a_3 s_1 s_2 s_3 - a_2 s_1 c_2 \\ -s_2 c_3 - c_2 s_3 & s_2 s_3 - c_2 c_3 & 0 & -a_3 s_2 c_3 - a_3 c_2 s_3 - a_2 s_2 + d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Viewing the last column:

$$\begin{bmatrix} a_3 c_1 c_2 c_3 - a_3 c_1 s_2 s_3 + a_2 c_1 c_2 \\ a_3 s_1 c_2 c_3 - a_3 s_1 s_2 s_3 + a_2 s_1 c_2 \\ -a_3 s_2 c_3 - a_3 c_2 s_3 - a_2 s_2 + d_1 \end{bmatrix} = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}$$

Gives:

$$x_c = a_3 c_1 c_2 c_3 - a_3 c_1 s_2 s_3 + a_2 c_1 c_2$$

$$y_c = a_3 s_1 c_2 c_3 - a_3 s_1 s_2 s_3 + a_2 s_1 c_2$$

$$z_c = -a_3 s_2 c_3 - a_3 c_2 s_3 - a_2 s_2 + d_1$$

The variables in this equations is θ_1 , θ_2 and θ_3 . Observed that deviding y_c by x_c will give an expression for θ_1 :

$$\frac{y_c}{x_c} = \frac{a_3 s_1 c_2 c_3 - a_3 s_1 s_2 s_3 + a_2 s_1 c_2}{a_3 c_1 c_2 c_3 - a_3 c_1 s_2 s_3 + a_2 c_1 c_2} = \frac{s_1 (a_3 c_2 c_3 - a_3 s_2 s_3 + a_2 c_2)}{c_1 (a_3 c_2 c_3 - a_3 s_2 s_3 + a_2 c_2)} = \frac{s_1}{c_1} = \tan(\theta_1)$$

$\text{atan2}(y, x) = \theta_1$

Trying to find expressions for θ_2 and θ_3 we can use the fact that $\sin(u + v) = \sin(u)\cos(v) + \cos(u)\sin(v)$ and $\cos(u + v) = \cos(u)\cos(v) - \sin(u)\sin(v)$:

$$x_c = a_3 c_1 (c_2 c_3 - s_2 s_3) + a_2 c_1 c_2 = a_3 c_1 c_{23} + a_2 c_1 c_2 = c_1 (a_3 c_{23} + a_2 c_2)$$

$$y_c = a_3 s_1 (c_2 c_3 - s_2 s_3) + a_2 s_1 c_2 = a_3 s_1 c_{23} + a_2 s_1 c_2 = s_1 (a_3 c_{23} + a_2 c_2)$$

$$z_c = -a_3 (s_2 c_3 + c_2 s_3) - a_2 s_2 + d_1 = -a_3 s_{23} - a_2 s_2 + d_1$$

If we square x_c , y_c and z_c we can try to use the identity $c^2 + s^2 = 1$. And if we move d_1 in the z_1 -expression, the calculations will be easier:

$$\begin{aligned}x_c^2 &= c_1^2(a_3c_{23} + a_2c_2)^2 \\y_c^2 &= s_1^2(a_3c_{23} + a_2c_2)^2 \\(z_c - d_1)^2 &= (-a_3s_{23} - a_2s_2)^2\end{aligned}$$

Adding x_c^2 and y_c^2 together:

$$x_c^2 + y_c^2 = c_1^2(a_3c_{23} + a_2c_2)^2 + s_1^2(a_3c_{23} + a_2c_2)^2 = (\cancel{c_1^2} + \cancel{s_1^2})(a_3c_{23} + a_2c_2)^2 = (a_3c_{23} + a_2c_2)^2$$

Adding in the $(z_c - d_1)^2$:

$$\begin{aligned}x_c^2 + y_c^2 + (z_c - d_1)^2 &= (a_3c_{23} + a_2c_2)^2 + (-a_3s_{23} - a_2s_2)^2 \\&= a_3^2c_{23}^2 + 2a_2a_3c_2c_{23} + a_2^2c_2^2 + a_3^2s_{23}^2 + 2a_2a_3s_2s_{23} + a_2^2s_2^2 \\&= a_3^2(\cancel{c_{23}^2} + \cancel{s_{23}^2}) + 2a_2a_3c_2c_{23} + 2a_2a_3s_2s_{23} + a_2^2(\cancel{c_2^2} + \cancel{s_2^2}) \\&= a_3^2 + 2a_2a_3c_2c_{23} + 2a_2a_3s_2s_{23} + a_2^2\end{aligned}$$

From here we can write c_{uv} out as $\cos(u)\cos(v) - \sin(u)\sin(v)$ and s_{uv} as $\sin(u)\cos(v) + \cos(u)\sin(v)$

$$\begin{aligned}&= a_3^2 + 2a_2a_3c_2(c_2c_3 - s_2s_3) + 2a_2a_3s_2(s_2c_3 + c_2s_3) + a_2^2 \\&= a_3^2 + 2a_2a_3c_2^2c_3 - \cancel{2a_2a_3c_2s_2s_3} + 2a_2a_3s_2^2c_3 + \cancel{2a_2a_3s_2c_2s_3} + a_2^2 \\&= a_3^2 + 2a_2a_3c_3(\cancel{c_2^2} + \cancel{s_2^2}) + a_2^2 \\&= a_3^2 + 2a_2a_3c_3 + a_2^2\end{aligned}$$

This expression has only one variable (θ_3), which gives:

$$\begin{aligned}x_c^2 + y_c^2 + (z_c - d_1)^2 &= a_3^2 + 2a_2a_3c_3 + a_2^2 \\2a_2a_3c_3 &= x_c^2 + y_c^2 + (z_c - d_1)^2 - a_3^2 - a_2^2 \\c_3 = \cos(\theta_3) &= \frac{x_c^2 + y_c^2 + (z_c - d_1)^2 - a_3^2 - a_2^2}{2a_2a_3} := D\end{aligned}$$

To find a singular and defined value for all θ_3 we use $\arctan2$ and expression D:

$$\begin{aligned}s_3 &= \pm\sqrt{1 - c_3^2} = \pm\sqrt{1 - D^2} \\ \tan(\theta_3) &= \frac{s_3}{c_3} = \frac{\pm\sqrt{1 - D^2}}{D} \\ \theta_3 &= \text{atan2}(\pm\sqrt{1 - D^2}, D)\end{aligned}$$

And then the only missing variable is θ_2 , so we try to make the x_c -expression on the form $a \cos \theta + b \sin \theta = c$ so that θ can be found by the formula $\theta = \text{atan2}(\frac{b}{a}) \pm \text{atan2}(\frac{\sqrt{a^2+b^2+c^2}}{c})$:

$$x_c = a_3 c_1 c_2 c_3 - a_3 c_1 s_2 s_3 + a_2 c_1 c_2$$

$$x_c = (a_3 c_1 c_3 + a_2 c_1) c_2 + (-a_3 c_1 s_3) s_2$$

To fill in the formula: $a = a_3 c_1 c_3 + a_2 c_1$, $b = -a_3 c_1 s_3$ and $c = x_c$:

$$\theta_2 = \frac{\text{atan2}(\frac{-a_3 c_1 s_3}{a_3 c_1 c_3 + a_2 c_1}) \pm \text{atan2}(\frac{\sqrt{(a_3 c_1 c_3 + a_2 c_1)^2 + (-a_3 c_1 s_3)^2 + x_c^2}}{x_c})}{}$$

4c)

Number of solutions for different positions of the tip (of link 3):

- **1 if the tip of link 3 is at the border of workspace (except special case*)**

Because to reach the outer border, link 3 must be an extension of link 2 (stretched out). The border of workspace can also be reached if link 2 and 3 are extensions of link 1, but this is a special case (defined under).

- **2 if the tip of link 3 is in the workspace (except special case*)**

θ_2 and θ_3 can vary by making the elbow down or elbow up.

- *** ∞ if the tip of link 3 intersects the z_0 -axis**

In this special case the angle of θ_1 does not matter. So it can rotate 365 degrees, and the tip of link 3 will still be at the same position.